AberdeenGroup
Perspective
Insights and Advice
Enterprise Integration Research
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Why Aberdeen Group is Focusing on Enterprise IT Integration

Business Challenge

The time has come for enterprise management, not just the CIO, to look at the "forest" as a whole, not be distracted by today's tactical "trees". There are megatrends in both information technology (IT) and business opportunities enabled by technology. Aberdeen's Enterprise IT Integration practice seeks to identify technology types and uses, which are synergistic: over time, IT investments end up being worth more than the sum of the individual projects. This Perspective examines the megatrends we are observing in 2006 and discusses the business-value themes that will follow our fact-based research in IT and in conjunction with other Aberdeen research practices.

Aberdeen Perspective

The Megatrends Driving Information Technology Integration Opportunity

Our perspective is based on almost two decades of watching the evolution of IT through market research. The key forces driving opportunities in information technology are:

1. **Programmer productivity** is stuck at the 1970s level at many shops. Legacy application spaghetti-code is legendary for its ability to soak up enormous talent for trivial gains. A change is needed because the world just does not have the programmers coding in 3GLs needed to solve the business problems at hand. It is telling that Microsoft – a bastion of technology brainpower – halted work on its Windows Vista OS and threw away tens of millions of lines of code in 2004. Why? To institute a programming process change that should enable a lower life-cycle cost. Aberdeen sees opportunities in service-oriented architectures, composite applications, and business process modeling as examples of development technology that can dramatically improve the quality, timeliness, and productivity of application developers and analysts. Are programmer productivity and program quality the weakest links in the IT chain? Our research shows best-in-class IT organizations freeing up to 15% of the IT budget with

Forces Triggering a Call For Change

Traditional approaches to building and deploying IT systems are not scaling with elegance. Brute force programming and enormous investments in hardware and software infrastructure are expensive to maintain and difficult to change. Yet Internet applications and flat-world globalization challenges are driving demands for flexibility, rapid IT change, grow-with-explosive-demand volumes, higher and smarter worker productivity, and collaboration with customers and supply partners in ways barely dreamed of a decade ago. Since IT budget growth cannot eat up the bottom line, and business-driven IT complexity is assured, enterprises need to take a more holistic, longer-term view on maximizing the value of IT personnel and infrastructure investments.

lower software maintenance costs compared to the average IT organization. Most of the savings are redeployed to IT innovation that drives immediate and tangible business value.

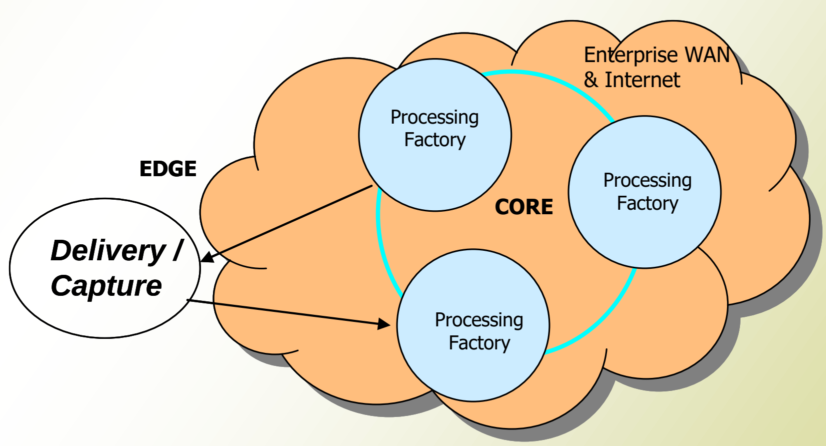
2. **Systems-Oriented Architecture and Web Services** based on industry standards are in use at many enterprises and will make up the majority of development by the turn of the decade. A lot of valuable solutions in this space from new and well-established suppliers will ease the transition. However, the real payback comes when existing "legacy" applications can be wrapped as an SOA-based service. Some ten years down the road, investments in SOA and reusable code will be paying back enormous dividends in lower IT application asset life-cycle costs and faster, more flexible application changes. Lastly, an SOA is a good vehicle for *virtualizing* the processing, storage, and network components of IT infrastructure (see below).

3. **Deconstructing Processing at the Core and Edge and on the Network** is driven by a new perspective on an old model, spoke-node-ring (SNR), Aberdeen's first framework in 1988. SNR foreshadowed the client-server and LAN/WAN revolution. Looking forward, we see the core datacenter "*factories*" deliberately breaking out processing (i.e., blades), storage (i.e., SANs) and datacenter fabric connecting LANs and WANs. Enterprise datacenters must be replicated and support data synchronization and fast switch-over on failure. 24-7 was seldom needed 20 years ago. Today, it is mandatory for most organizations.

   The *edge* is where data are delivered to people and devices, and from which data are captured. The edge is gaining lots of intelligence, as exemplified by the growth in retail store and shop floor computing power. The edge is an opportunity for value generation with careful integration.

   Sun Microsystems' trademark is "the network is the computer". Watch out, Sun and other server suppliers, because true networking vendors such as Cisco are angling to provide intelligent network service *networks* that to date have been run on conventional servers. Our perspective is that solutions selection will become more complex with the advent of network processing but also more competitive, as new approaches will prove desirable.

Figure 1: Aberdeen Group's Enterprise IT Integration Model



Source: AberdeenGroup, February 2006

Figure 1 depicts our Enterprise Integration model, which is composed of:

- *Edge* components consisting of fixed and mobile people and devices such as laptops and RFID stations. The edge is where data are first captured, and the end point for delivery. The edge will often be connected via the Internet, so secure authentication, encryption, and all the Internet security concerns and solutions apply.
- *Networking* consists of Internet and enterprise WAN components. What we see changing are two factors: first, the network itself supplies more computing services such as encryption, compression, queuing and messaging, quality of service, and other activities that today are performed by servers attached to the network. The enterprise WAN and related managed services may be an outsourcing candidate. Edge-attached appliances also have a role, as there are many benefits to moving certain processing and data closer to the edge user. The business value is in optimizing the security, performance, and reliability of the WAN backbone and its operations.
- *Processing Factories* are deconstructed datacenters that will gradually become "JBOR" (just a bunch of racks). Traditional computers are already splitting into computing servers and Storage Area Networks. Datacenter fabrics are adequate to tie together the datacenter LAN segments. We anticipate further server consolidation as virtualization technology eases the complexity of running and isolating multiple workloads. The monolithic enterprise datacenter is, fortunately, seldom employed today.

  Chip-maker focus on instructions per watt will allow greater density, as will better cooling technology, lowering HVAC demands. Operational tools to remotely monitor, virtually reconfigure, and deploy servers are needed to keep labor costs under control.

  Leading companies will translate lessons from scientific massively parallel computing into a grid for commercial computing. The insight is the movement towards greater real-time operational analysis and decision making (see below).

  The business goal of enterprise integration is to achieve a *smart processing* architecture, in which "where" and "how" computing is done is continually optimized for business process conditions – without throwing out today's IT investments.

4. **Mobility is the Norm, Not the Exception.** There are many applications running today that were designed in a business world where immobile terminals and devices on land lines connected to computers with a very fixed view of the world. Mobility technology is opening up new business-generation and service opportunities while enabling new employee work models such as telecommuting and ad hoc meeting collaboration. Driving business value requires integrating a mobility infrastructure with the existing static infrastructure – with the added security and performance issues inherent to mobility technology. Our point of view is that mobility technology is technically mature enough in 2006, but that many enterprises have only a tepid, reactive strategy for generating business value. We will examine the market's maturity, best practices and solution selection.

5. **Real-Time Business Intelligence** (RTBI) demands the rapid consolidation of business-critical data to drive key operational performance metrics. The mostly large companies which have RTBI experience consider the business value generation so critical to success that the "how" becomes a company secret – a competitive weapon. Data warehouses and datamarts traditionally look at older information, meaning an insight into where the business was not where it is or will be. Our perspective is that the gradual "replumbing" of IT infrastructure with a service-driven architecture creates new opportunities to instrument enterprise IT management (e.g., transaction response times and server outages) as well as new business process management instrumentation that can drive much faster, "near real time", universal, and lower level decision-making. However, there are significant issues in data management, quality, timeliness, and processing. Enterprise information integration (EII) is technology that promises to augment traditional data warehouse technology. Nevertheless, we view RTBI as a megatrend offering the promise of significant business benefits.
6. **Knowledge Worker (KW) Productivity** is an attractive tangential target due to the requirement to manage mobility and the desirability to drive better RTBI, discussed above. Other technology vectors contributing to KW productivity include collaboration, messaging, smart phones, laptops, podcasting, conferencing, enterprise portals, project management and compliance, and improvements in enterprise applications such as Microsoft Office, ERP, sales force automation, and customer relationship management. Integrating this technology with the trends discussed above is a rich source of value generation. For example, best-in-class organizations show positive ROI for laptops merely with a few minutes of e-mail time gained while waiting for meetings to start.

Bottom Line: Act Tactically but Build Strategically

No reader can start with a clean sheet of paper and no legacy. The reality is that there is a legacy and new investments are limited. We posit the need for a focus on synergy: over time, IT investments end up being worth more than the sum of the individual projects. Our experience is that the integration of information technologies and the integration of IT with business processes are areas ripe for improving value: efficiency, speed, productivity, useful life, and quality.

Our perspective is that the tough but best approach is to have a strategic IT architecture that incorporates the megatrends discussed above, and to use every tactical project opportunity to get a step closer to making the development process, edge and datacenter infrastructure, mobility, and data management intelligence-generation capabilities the big contributors to business value generation.

The Enterprise Integration practice continually researches the market, assisting our IT management readers in benchmarking their organization's maturity, examining best practices, and studying the lessons learned from technology solution selection and implementation. We invite IT managers to participate in our fact-based research studies and to use the CIO portal at Aberdeen.com.

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Related Research

The SOA in IT Benchmark Report, December 2005

For Mid-Size Enterprises, SOA's Benefits Begin with IT, January 2006

The Service-Oriented Architecture in the Supply Chain Benchmark Report, October 2005

SOA Success Starts with IT Success, November 2005

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